Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Negotiate, Then Solve: Agentic AI for Scheduling | *Python, PyTorch, LLM Agents, CP-SAT, FastAPI*

- Built an AI agent that drafts schedule changes, checks them with a solver and negotiates with the people affected.
- Fine-tuned a small 4B model with LoRA on the agent's traces; it **outperformed the Gemini model** it learned from.
- Compared the agent with a fixed pipeline on cost, reliability and recovery from mistakes.
- Added safety checks (authority, cited rules, human approval); **no attack succeeded** in 50 prompt-injection tests.

Retrieve Ten, Keep Three: Re-ranking RAG Evaluation | *Python, FAISS, BM25, FastAPI*

- Built a question-answering system over the GDPR that retrieves 10 passages and re-ranks them to keep the best 3.
- Re-ranking raised top-3 retrieval accuracy from **46% to 85%**; combining keyword and embedding search helped further.
- Built an LLM judge checked against hand labels, and found many "wrong" answers came from mislabelled test data.
- Deployed it as a web app where users upload PDFs and compare search methods side by side.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, SentencePiece, ONNX Runtime Web*

- Pretrained GPT-style language models for Hindi and Assamese from scratch in PyTorch, including the data and tokenizers.
- Fine-tuned them for multi-step reasoning, reaching **96%** test accuracy against a 48% baseline.
- Exported the models to run fully in the browser and deployed them as a live demo.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style file sharing system in C++ with a central tracker, multithreaded peers and per-piece integrity checks.
- Profiled and fixed bottlenecks in scheduling and networking, cutting a 1GB download from **193 s to under 13 s**.
- Added a replicated backup tracker with automatic failover, tested with unit tests under sanitizers.

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL.
AI & LLMs: LLM APIs (Gemini), LoRA, RAG, FAISS, llama.cpp.
Machine Learning: PyTorch, Hugging Face, Scikit-learn, NumPy, ONNX.
Tools: FastAPI, React, pytest, Git, Docker, Linux.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.